Cones over Forests and the Chromatic-Polynomial Graphon Bound

Abstract

Let F be a forest with n vertices, $e \geq 1$ edges, and c connected components, so $n = c + e$. We prove that its cone $H = K_1 \vee F$ satisfies

$$t(H, W) \geq p^c(2p-1)^e = \Phi_H(p)$$

for every graphon W of edge density $p \geq 1/2$. Since $\chi(H) = 3$, this is the complete required density interval. The proof uses the accepted Sidorenko inequality for forests inside the graphon links and a fully proved weighted rooted-triangle estimate. It is a special forest-cone theorem, not a general universal-vertex or join closure. In the catalogue on at most six vertices it proves the previously open Atlas graphs 40, 135, and 136.

1 Definitions, chromatic polynomial, and target

Let (Ω, μ) be a probability space. A graphon is a symmetric measurable function $W\colon \Omega^2 \to [0,1]$. For a finite simple graph G, write

$$t(G, W) = \int_{\Omega^{V(G)}} \prod_{uv \in E(G)} W(x_u, x_v) \prod_{u \in V(G)} \mathrm{d}\mu(x_u).$$

This is the ordinary, non-injective homomorphism density. Put

$$p = t(K_2, W), \qquad d(x) = \int_\Omega W(x, y)\, \mathrm{d}\mu(y).$$

Let F be a forest with n vertices, $e \geq 1$ edges, and c connected components, including isolated vertices as one-vertex components. Each tree component on v_i vertices has chromatic polynomial $x(x-1)^{v_i - 1}$. Therefore

$$\chi_F(x) = x^c(x-1)^e, \qquad n = c + e.$$

Define the cone

$$H = K_1 \vee F$$

by adding a new vertex o adjacent to every vertex of F. Colouring o first gives

$$\chi_H(x) = x\chi_F(x-1) = x(x-1)^c(x-2)^e. \tag{1}$$

Because F has an edge, H contains a triangle and is 3-colourable; hence $\chi(H) = 3$.

Let $q = 1 - p$. Substitution in the definition of the target yields

$$\begin{aligned} \Phi_H(p) &= q^{n+1}\chi_H\left(\frac{1}{q}\right) \\ &= q^{n+1}\frac{1}{q}\left(\frac{p}{q}\right)^c\left(\frac{2p-1}{q}\right)^e \\ &= p^c(2p-1)^e, \end{aligned} \tag{2}$$

because $n = c + e$. The identity holds at $p = 1$ by continuous extension.

2 The accepted forest input

We use the following standard, previously accepted positive result.

Theorem 2.1 (Sidorenko inequality for forests)**.** *Let F be a forest with e edges. Every graphon V of edge density z satisfies*

$$t(F, V) \geq z^e.$$

Isolated vertices make no change to homomorphism density, disjoint unions multiply, and the connected case is the classical tree Sidorenko inequality. This theorem is the only previously accepted graph-density result used below.

3 Weighted density in graphon links

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x, y)W(x, z)W(y, z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Lemma 3.1 (Weighted rooted-triangle inequality)**.** *If $p > 0$, then for every integer $h \geq 0$,*

$$\int_\Omega d(x)^h\tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p}\int_\Omega d(x)^{h+2}\,\mathrm{d}\mu(x).$$

Proof. For $j \geq 0$, put $M_j = \int_\Omega d(x)^j\,\mathrm{d}\mu(x)$. Let T_W be the integral operator with kernel W, and define

$$I_h = \int_\Omega d(x)^h\tau(x)\,\mathrm{d}\mu(x), \qquad J_h = \left\langle d^h, T_W d\right\rangle.$$

For $a, b \in [0, 1]$, one has $ab \geq a + b - 1$. Apply this to $W(x, y)$ and $W(x, z)$, multiply by the nonnegative quantity $d(x)^h W(y, z)$, and integrate. This gives

$$I_h \geq 2J_h - pM_h. \tag{3}$$

Set $U = 1 - W$, let T_U be its integral operator, and put

$$u = T_U\mathbf{1} = 1 - d, \qquad q = 1 - p.$$

Since $T_U d = T_U(\mathbf{1} - u) = u - T_U u$, we have

$$T_W d = p\mathbf{1} - T_U d = d - q\mathbf{1} + T_U u.$$

Consequently,

$$J_h = M_{h+1} - qM_h + \left\langle d^h, T_U u\right\rangle. \tag{4}$$

By symmetry of U, twice the difference

$$\left\langle d^h, T_U u\right\rangle - \int_\Omega d(x)^h u(x)^2\,\mathrm{d}\mu(x)$$

equals

$$\int_{\Omega^2} U(x, y)\big(u(y) - u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y).$$

This is nonnegative, because $d = 1 - u$, so u and d^h are oppositely ordered. Hence (4) implies

$$\begin{aligned} J_h &\geq M_{h+1} - qM_h + \int d^h(1-d)^2 \\ &= M_{h+2} - M_{h+1} + pM_h. \end{aligned}$$

Using this in (3),

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

Finally,

$$\begin{aligned} &2M_{h+2} - 2M_{h+1} + pM_h - \frac{2p-1}{p} M_{h+2} \\ &\quad = \frac{1}{p} \int_\Omega d(x)^h (d(x) - p)^2 \, \mathrm{d}\mu(x) \geq 0. \end{aligned}$$

□

For $d(x) > 0$, define a probability measure

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)} \, \mathrm{d}\mu(y)$$

and let W_x denote the same kernel W on this probability space. Its edge density is

$$z_x = t(K_2, W_x) = \frac{\tau(x)}{d(x)^2}.$$

If $d(x) = 0$, set $z_x = 0$.

Corollary 3.2 (Weighted average link density)**.** *For every integer $N \geq 2$ and every $p > 0$,*

$$\frac{\int_\Omega d(x)^N z_x \, \mathrm{d}\mu(x)}{\int_\Omega d(x)^N \, \mathrm{d}\mu(x)} \geq 2 - \frac{1}{p}.$$

Proof. The numerator is $\int d^{N-2}\tau$. Apply Lemma 3.1 with $h = N - 2$. The denominator is positive because $p > 0$. □

4 The forest-cone theorem

Theorem 4.1 (Cones over forests)**.** *Let F be a forest with n vertices, $e \geq 1$ edges, and c connected components. Let $H = K_1 \vee F$. Every graphon W with edge density $p \geq 1/2$ satisfies*

$$\boxed{t(H, W) \geq p^c (2p-1)^e = \Phi_H(p).}$$

Since $\chi(H) = 3$, the interval $p \geq 1/2$ is exactly the full required interval $p \geq 1 - 1/(\chi(H) - 1)$.

Proof. Condition on the image x of the cone vertex o. Directly from the homomorphism integral,

$$t(H, W) = \int_\Omega d(x)^n t(F, W_x) \, \mathrm{d}\mu(x). \tag{5}$$

At points where $d(x) = 0$, both the original conditional integral and the displayed integrand vanish. Thus no pointwise positive-degree assumption is being made.

By Theorem 2.1,

$$t(F, W_x) \geq z_x^e.$$

Put $M_n = \int d^n$. Since $e \geq 1$, the function $z \mapsto z^e$ is convex and nondecreasing on $[0, 1]$. Jensen's inequality with respect to the probability measure $d(x)^n \, \mathrm{d}\mu(x)/M_n$, followed by Corollary 3.2, gives

$$\begin{aligned} t(H, W) &\geq \int_\Omega d(x)^n z_x^e \, \mathrm{d}\mu(x) \\ &\geq M_n \left(\frac{\int d(x)^n z_x \, \mathrm{d}\mu(x)}{M_n} \right)^e \\ &\geq M_n \left(2 - \frac{1}{p} \right)^e. \end{aligned}$$

The last step uses $p \geq 1/2$, so the lower bound on the weighted mean is nonnegative. A second application of Jensen gives $M_n \geq p^n$. Therefore

$$\begin{aligned} t(H, W) &\geq p^n \left(\frac{2p-1}{p} \right)^e \\ &= p^{n-e}(2p-1)^e = p^c(2p-1)^e. \end{aligned}$$

This is exactly (2). □

5 Endpoints and equality examples

At $p = 1/2$, the target is zero because $e \geq 1$, so the conclusion also follows directly from nonnegativity. The balanced complete bipartite graphon has no triangle and attains equality. At $p = 1$, one has $W = 1$ almost everywhere, and both sides equal 1.

For every integer $k \geq 2$, the balanced complete k-partite graphon T_k has edge density $p = 1 - 1/k$ and satisfies

$$t(H, T_k) = \frac{\chi_H(k)}{k^{v(H)}} = \Phi_H(p).$$

Thus the bound has the expected equality examples at every multipartite grid point in the required interval.

6 Catalogue consequences

Corollary 6.1 (New positive Atlas cases)**.** *The following previously open graphs satisfy the desired inequality for every graphon throughout the full interval $p \geq 1/2$:*

Atlas ID	*graph6*	(c, e) *for* F	$\chi_H(x)$	$\Phi_H(p)$
40	`DjW`	$(2, 2)$	$x(x-1)^2(x-2)^2$	$p^2(2p-1)^2$
135	`EzW_`	$(2, 3)$	$x(x-1)^2(x-2)^3$	$p^2(2p-1)^3$
136	`Ejt?`	$(2, 3)$	$x(x-1)^2(x-2)^3$	$p^2(2p-1)^3$

Proof. Deleting the universal vertex from Atlas 40 leaves a three-vertex path and one isolated vertex, so $(c, e) = (2, 2)$. Deleting the universal vertex from either Atlas 135 or Atlas 136 leaves a four-vertex tree and one isolated vertex, so $(c, e) = (2, 3)$. The two four-vertex trees are nonisomorphic, but Theorem 4.1 applies to every forest and depends only on c and e. The graph6 codes are independent machine-readable identifiers. □

7 Scope and accepted input

The only previously accepted graph-density result used is the forest Sidorenko inequality, Theorem 2.1. The weighted link estimate is proved here in full.

The theorem does not say that $K_1 \vee G$ is positive whenever G is positive. Its proof depends specifically on the forest inequality $t(F,V) \geq z^{e(F)}$ and on the convexity of the monomial z^e. Consequently, it establishes exactly cones over forests and no general universal-vertex or arbitrary-join closure.

The argument applies directly to arbitrary graphons and ordinary homomorphism densities. It uses no finite-step approximation, regularity, injectivity, minimum-degree assumption, or pointwise lower bound on the degree function.

A What the Lean formalization actually proves

The development is `lean/Taeyoung/Methods/ForestCone/Rows.lean`, over `Methods/ConeBound.lean`. All five Atlas rows this methodology is credited with — 40, 111, 117, 135, 136 — are `verified`, as is 192, the cone over 40. One difference of substance.

Theorem 2.1 is never assumed

The note takes the forest Sidorenko inequality as an accepted input. Nothing in the catalogue may rest on an accepted input, so Theorem 4.1 is formalized in the conditional form `satisfiesLowerBound_coneGraph_of_sidorenko`, with $t(F,V) \geq z^e$ a hypothesis, and that hypothesis is discharged separately for each row. Four of the five rows turn out not to need anything of the kind: deleting the universal vertex leaves

Atlas	base F	why $t(F,V) \geq z^{e(F)}$
40	$K_{1,2} \sqcup K_1$	$t(K_{1,n},V) = M_n \geq z^n$, Jensen
111	$K_{1,2} \sqcup 2K_1$	the same
117	$K_2 \sqcup K_2 \sqcup K_1$	multiplicativity, with equality
135	$K_{1,3} \sqcup K_1$	Jensen again
136	$P_4 \sqcup K_1$	**not** a star: needs $t(P_4,V) \geq z^3$

so only Atlas 136 uses a genuine tree inequality, and only the single case P_4. That case is proved outright in `Methods/PathSidorenko.lean`, by three-factor Hölder on the almost-everywhere factorization $W = (Wdd')^{1/3}(W/d)^{1/3}(W/d')^{1/3}$. Tree Sidorenko in general, and hence Theorem 2.1 in general, is not formalized and is not needed.

Lemma 3.1 is formalized as stated, and is shared with the clique common-leaf and cone families.